Conditional probability-based email SPAM filter

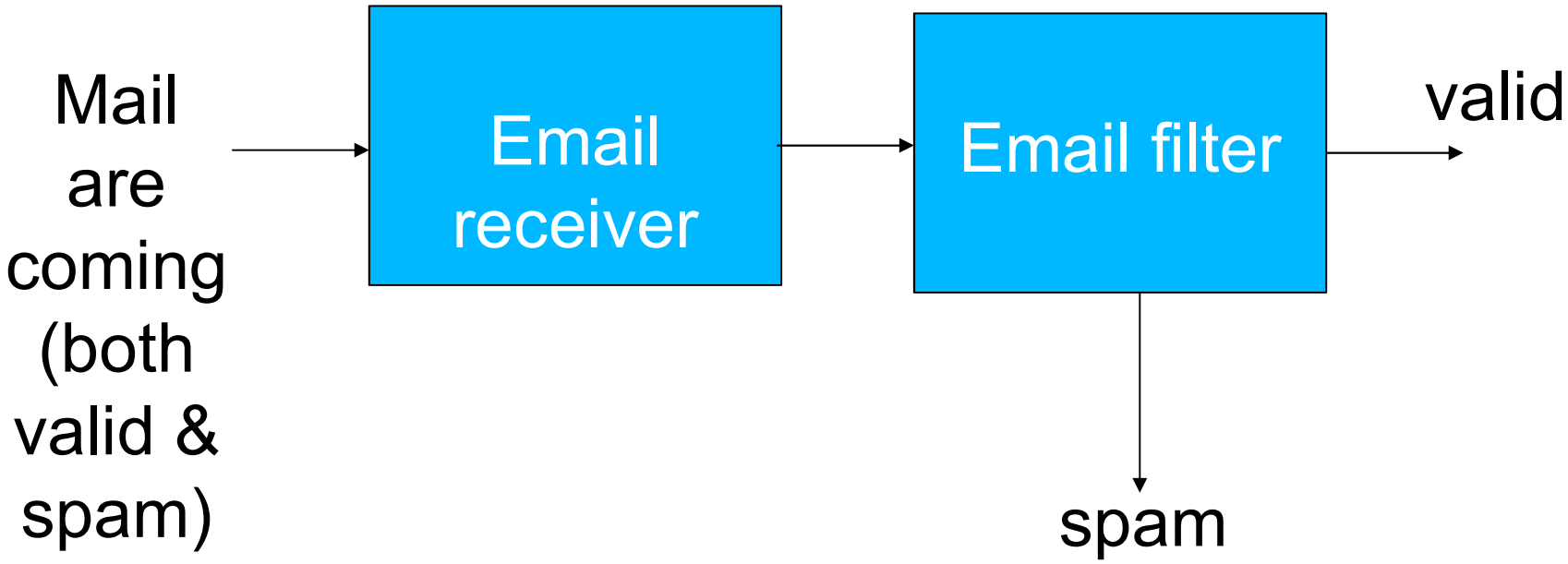
e.g.

<u>Email spam filter</u>

- We receive valid and spam mails
- Design a spam filter

Observation

- After designing validate your design by observing


Mail are coming (both valid & spam)
Email receiver
Email filter
valid
spam

e.g.

<u>Email spam filter</u>

- We receive valid and spam mails
- P(spam) = 2%
- Spam detector detects the spam mails
- But it has false positive problem
- P(detected as spam|spam) = 98%
- P(detected as spam|Not-spam) = 0.2%

$$P(spam \mid detected\ as\ spam)$$
$$= \frac{P(detected\ as\ spam \mid spam).P(spam)}{P(detected\ as\ spam)}$$

- Both good as well as spam mails can be detected as spam

$$P(detected\ as\ spam)$$
$$= P(detected\ as\ spam \cap spam)$$
$$+ P(detected\ as\ spam \cap NotSpam)$$

$$P(detected\ as\ spam \cap spam)$$
$$= P(detected\ as\ spam|spam)P(spam)$$
$$= 0.99 * 0.02 = 0.0198$$
$$P(detected\ as\ spam \cap NotSpam)$$
$$= P(detected\ as\ spam|NotSpam)P(NotSpa$$
$$= 0.002 * 0.98 = 0.00196$$
$$P(detected\ as\ spam) = 0.0198 + 0.00196$$
$$= 0.02176$$

$$P(spam \mid detected\ as\ spam)$$

$$= \frac{P(detected\ as\ spam \mid spam).P(spam)}{P(detected\ as\ spam)}$$

$$= \frac{(0.98 * 0.02)}{0.02176} = 0.9 = 90\%$$

$$P(NotSpam \mid detected\ as\ spam)$$
$$= \frac{P(detected\ as\ spam \mid NotSpam).P(NotSpam)}{P(detected\ as\ spam)}$$

- Both good as well as spam mails can be detected as spam

$$P(detected\ as\ spam)$$
$$= P(detected\ as\ spam \cap spam)$$
$$+ P(detected\ as\ spam \cap NotSpam)$$

$$P(detected\ as\ spam \cap spam)$$
$$= P(detected\ as\ spam|spam)P(spam)$$
$$= 0.99 * 0.02 = 0.0198$$
$$P(detected\ as\ spam \cap NotSpam)$$
$$= P(detected\ as\ spam|NotSpam)P(NotSpa$$
$$= 0.002 * 0.98 = 0.00196$$
$$P(detected\ as\ spam) = 0.0198 + 0.00196$$
$$= 0.02176$$

$$P(detected\ as\ spam|NotSpam)$$

$$= \frac{P(detected\ as\ spam \cap NotSpam)}{P(NotSpam)} = \frac{0.002}{0.98}$$

$$= 0.002$$

$$P(NotSpam\ |detected\ as\ spam)$$

$$= \frac{P(detected\ as\ spam|NotSpam).P(NotSpam)}{P(detected\ as\ spam)}$$

$$= \frac{0.002 * 0.98}{0.02176} = 0.09 = 9\%$$

$$P(NotSpam \mid detected\ as\ spam) = 9\%$$
$$P(Spam \mid detected\ as\ spam) = 90\%$$